
Fig. 12: P.Eye Raman explosives detector

Fig. 15: ReporteR Raman detector

Fig. 13: ObserveR Raman detector

Fig. 16: THOR-1064 explosive detector

Fig. 14: Inspector-300 Raman detector


Fig. 17: Preemptor by LASTEC (DRDO)

rate. It derives power from UH-60 helicopter platform.

LR BSDS is intended to be a corps-level asset that provides early warning and information on the location and configuration of aerosol cloud formation. This information is used to enhance contamination avoidance efforts and cue other biological detection measures. Information from LR BSDS is analysed with other battlefield information and intelligence data to determine appropriate defensive measures.

Laser-based detection of explosive agents

Stand-off detection of explosive agents using laser-based trace detection methods is one of the most widely researched technologies internationally. For the technology to mature to an extent where it can be transformed into a product usable in the kind of environment or field condition usually encountered in homeland security-related applications is a great technological challenge.

One of the major problems comes from decrease in intensity of backscattered light signal due to wavelength-dependent absorption and scattering losses, and intensity decreasing inversely with distance squared. The problem is compounded by the fact that trace levels associated with common explosive agents are extremely low (in the range of fraction of parts per billion to few parts per million) for common explosives.

Another major problem pertains to unique identification of targeted explosive agents in a background of interferents. Many chemical agents have atomic compositions including sulphur, phosphorus, fluorine and chlorine, in addition to nitrogen, oxygen, hydrogen and carbon present in organic molecules. Thus, detection methodology needs to be highly sensitive and selective. Laser-based spectrometric methods have the potential of being fast, sensitive and selective with the ability to detect and identify a wide range of explosive agents, and to upgrade to handle new threats.

Atmospheric transmission at wavelengths concerned is an important factor while assessing suitability of a given stand-off detection methodology. Commonly-used technologies are laser-induced breakdown spectroscopy (LIBS), Raman spectroscopy and its variants, LIF spectroscopy and IR spectroscopy. These are all trace-detection methods; although, bulk-detection methods such as millimetre-wave imaging and terahertz spectroscopy exist.

LIBS. This focuses a high-energy laser beam on the trace sample to break down a small part of the sample into plasma of excited ions and atoms. Plasma emits light that is characteristic of emissions from ionic, atomic and small molecular species. These light emissions are detected by a spectrometer to identify the elemental composition. Fig. 9 shows a typical LIBS setup.

One of the challenges in using LIBS is the need to assess its efficacy to detect and identify explosive species in real environment, which is replete with many interfering substances. One way to get the desired selectivity is to use double-pulse LIBS. In this, first pulse creates the so-called laser-generated vacuum. The second pulse transmitted a few microseconds later generates the return signal. Double-pulse LIBS also improves sensitivity, in addition to enhancing selectivity. Selectivity can be further improved by adding temporal resolution to LIBS emission analysis.

Wavelength of 1064nm has been widely employed for LIBS systems. Due to serious eye hazards posed by 1064nm, scientists have tried 266nm. This also allows the designer to build Raman capability into the system. 266nm has 600 times higher maximum permissible exposure limit as compared to 1064nm.

Lawrence Berkeley National Laboratory of Department of Energy in the US has done pioneering work in developing laser ablation technologies such as LIBS for the detection of explosives and other hazardous chemicals. Applied Spectra founded by one of the scientists of Berkeley lab developed a portable LIBS-based explosive detection system.

A military prototype field version of the system (Fig. 10) was reportedly field tested on more than 100 samples in 2008. The detector was able to discriminate with 85 per cent accuracy whether tested samples contained residues of several types of explosives from stand-off distances between 30 to 50 metres, or whether composition was of materials such as rock, wood, metal and plastics.

Raman spectroscopy. This offers another method for stand-off detection of explosive agents. It has been extensively used for many years as a standard analytical tool for the identification of chemical agents in the laboratory environment. Basis of detection in this case is the shift in wavelength caused by inelastic Raman scattering